# Chapter 10

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10.1. Let $\alpha_n : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ be the injection of abelian groups given by $\alpha_n(1) = p^{n-1}$, and let $\alpha : A \to B$ be the direct sum of all the $\alpha_n$. Show that the $p$-adic completion of $A$ is just $A$ but that the completion of $A$ for the topology induced from the $p$-adic topology of $B$ is the direct *product* of the $\mathbb{Z}/p\mathbb{Z}$. Deduce that $p$-adic completion is *not* a right-exact functor on the category of all $\mathbb{Z}$-modules.

10.2. In (Ex. 1), let $A_n = \alpha^{-1}(p^n B)$, and consider the exact sequence

$$0 \to A_n \to A \to A/A_n \to 0.$$

Show that $\varinjlim$ is not right exact, and compute $\varinjlim^1 A_n$.

10.3. Let $A$ be a Noetherian ring, $\mathfrak{a}$ an ideal, and $M$ a finitely generated $A$-module. Using Krull's Theorem and (Ex. 3.14), prove that

$$\bigcap_{n=1}^{\infty} \mathfrak{a}^n M = \bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}}),$$

where $\mathfrak{m}$ runs over all maximal ideals containing $\mathfrak{a}$.

Deduce that

$$\widehat{M} = 0 \iff \operatorname{Supp}(M) \cap V(\mathfrak{a}) = \emptyset.$$

*Proof.* By Krull's Theorem, it suffices to show that $x \in \bigcap_{\mathfrak{m}\supset\mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$ if and only if $x$ is annihilated by some element of $1+\mathfrak{a}$. We have $x \in \ker(M \to M_{\mathfrak{m}})$ if and only if $(x)_{\mathfrak{m}} = 0$. Then, by (Ex. 3.14), $x \in \bigcap_{\mathfrak{m}\supset\mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$, implies $\mathfrak{a} = \mathfrak{a}(x)$. (2.5) gives us that $x$ is annihilated by some element of $1+\mathfrak{a}$, as desired.

Conversely, suppose there is $\alpha \in \mathfrak{a}$ with $(1+\alpha)x = 0$. Then $(x)/\mathfrak{a}(x) = 0$ implies $(x)_{\mathfrak{m}}/\mathfrak{a}_{\mathfrak{m}}(x)\mathfrak{m} = 0$ for maximal ideals $\mathfrak{m} \supset \mathfrak{a}$, whence $(x)_{\mathfrak{m}} = 0$ for all $\mathfrak{m} \supset \mathfrak{a}$ by Nakayama's lemma. Thus, $x \in \bigcap_{\mathfrak{m}\supset\mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$, as desired.

By the above,

$$\begin{aligned}\ker(M \to \widehat{M}) &= \bigcap_{n=1}^{\infty} \mathfrak{a}^n M \\ &= \bigcap_{\mathfrak{m}\supset\mathfrak{a}} \ker(M \to M_{\mathfrak{m}}).\end{aligned}$$

For any $\mathfrak{p} \in V(\mathfrak{a})$, we have $\mathfrak{a} \subset \mathfrak{p} \subset \mathfrak{m}$ for some $\mathfrak{m}$ maximal. Then $A - \mathfrak{m} \subset A - \mathfrak{p}$ gives us

$$\ker(M \to M_{\mathfrak{m}} \to M_{\mathfrak{p}}) \supset \ker(M \to M_{\mathfrak{m}}),$$

whence

$$\ker(M \to \widehat{M}) = \bigcap_{\mathfrak{p}\subset\mathfrak{a}} \ker(M \to M_{\mathfrak{p}}).$$

Thus, $\operatorname{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$ if and only if $M = \ker(M \to \widehat{M})$. Clearly $\widehat{M} = 0$ is a sufficient condition. Conversely, if every constant sequence converges to 0, then every every sequence converges to 0, i.e. $\widehat{M} = 0$. □

10.5. Let $\mathfrak{a}$ be a Noetherian ring, and let $\mathfrak{a}$, $\mathfrak{b}$ be ideals in $A$. If $M$ is any $A$-module, let $M^{\mathfrak{a}}$, $M^{\mathfrak{b}}$ denote its $\mathfrak{a}$-adic and $\mathfrak{b}$-adic completions respectively. If $M$ is finitely generated, prove that $(M^{\mathfrak{a}})^{\mathfrak{b}} \cong M^{\mathfrak{a}+\mathfrak{b}}$.

10.6. Let $A$ be a Noetherian ring and $\mathfrak{a}$ an ideal in $A$. Prove that $\mathfrak{a}$ is contained in the Jacobson radical of $A$ if and only if every maximal ideal of $A$ is closed for the $\mathfrak{a}$-topology. A Noetherian topological ring in which the topology is defined by an ideal in the Jacobson radical is called a *Zariski ring*. Examples are local rings and, by (10.15d), $\mathfrak{a}$-adic completions.

*Proof.* Suppose that $\mathfrak{m}$ is a maximal ideal of $A$ containing $\mathfrak{a}$. Then $\mathfrak{m} \supset G_0 = \mathfrak{a}$ is a neighborhood of 0, so

$$A \setminus \mathfrak{m} = \bigcup_{f \in A \setminus \mathfrak{m}} f + \mathfrak{m}$$

is open in the $\mathfrak{a}$-topology, whence $\mathfrak{m}$ is closed.

Conversely, suppose $\mathfrak{m}$ is closed in the $\mathfrak{a}$-topology, and ta Then $1 + \mathfrak{m} \not\ni 0$ is closed, so $A \setminus (1 + \mathfrak{m})$ is a neighborhood of 0. Thus,

$$A \setminus \mathfrak{m} = \bigcup_{f \in A \setminus \mathfrak{m}} f + \mathfrak{m}$$

is a neighborhood of 0. Thus, $\mathfrak{a}^n \subset 1 + (A \setminus \mathfrak{m})$ for some $n > 0$. □

10.7. Let $A$ be a Noetherian ring, $\mathfrak{a}$ an ideal of $A$, and $\widehat{A}$ the $\mathfrak{a}$-adic completion. Prove that $\widehat{A}$ is faithfully flat over $A$ if and only if $A$ is a Zariski ring (for the $\mathfrak{a}$-topology).

*Proof.* By (10.14), $\widehat{A}$ is flat over $A$. By (10.13), $\widehat{A} \otimes M \cong \widehat{M}$. Then it suffices by (3.16) to show that $M \to \widehat{M}$ is injective for all finitely gnerated $M$ if and only if $A$ is Zariski. If $A$ is Zariski, then $M \hookrightarrow \widehat{M}$ is injective by (10.19).

Conversely, suppose that $M \hookrightarrow \widehat{M}$ is injective for all finitely generated $M$. By (Ex. 10.6), it suffices to show Let $\mathfrak{R} = \bigcap \mathfrak{m}$ denote the Jacobson radical of $A$, and $\mathfrak{R}'$ the Jacobson radical of $\widehat{A}$. For each maximal ideal $\mathfrak{m} \trianglelefteq A$, $\widehat{\mathfrak{m}} = \mathfrak{m}^e$ is Then

$$\begin{aligned} \mathfrak{a} &= \mathfrak{a}^{ec} \\ &= \widehat{\mathfrak{a}}^c \\ &\subset \end{aligned}$$

(Ex. 3.16a) and (10.15) □